Interpreting Order Effects

Quantum Cognition for Human-AI Decision Teaming

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M2S1

Order Differences and QQ

Which statistic compares F across orders?

Which statistic compares A/B agreement?

What records does each require?

An Imposed Answer-Pair Table

Answer pair	AB	BA
A yes, B yes	30	45
A yes, B no	10	25
A no, B yes	20	5
A no, B no	40	25

Both columns contain 100 illustrative interviews.

Agreement Shares and Order Rates

Calculate each order’s agreement share.

Compare A’s yes rate across orders.

State what is missing for an F comparison.

A Classical Sequential Explanation

Inspect the transition rules in the notebook.

The rules vary with which question comes first.

Which extra prediction would test this explanation?

Intervals Against a Practical Range

Case	Illustrative 95% interval	Decision relative to ±5 pp
A	−2 to +3 pp	Within the range
B	−12 to +14 pp	Unresolved
C	+1 to +4 pp	Within the range

Use a prespecified tolerance of ±5 percentage points.

The Resampling Unit

The project links six interviews to each persona.

Keep the complete block together.

Explain what independent sampling is assumed.

Design Revisions

Freeze A, B, F, coding, and collection conditions.

Choose a practical threshold and q tolerance.

Write a fair test of the strongest alternative.

The Interpretation Paragraph

Explain why order and QQ answer different questions.

Describe what an uncertain interval leaves unresolved.

Use that reasoning in Discussion 2.